

CHARLOTTE LANGE

Machine Learning for Climate

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🌐 homepage

EDUCATION

M.Sc Environmental Science

Universität Freiburg

📅 October 2024 –

📍 Freiburg, DE

In the *Environmental Modelling and Data Science* major, I focus on machine-learning applications to environmental challenges, alongside coursework in statistics, environmental data wrangling and ecological systems.

1-year ELIZA scholarship (Konrad Zuse School of Excellence).

B.Sc. Cognitive Science

Universität Osnabrück

📅 October 2018 – April 2023 📍 Osnabrück, DE

In this highly interdisciplinary program, my personal focus lay on Deep Learning, Machine Learning, Neuroinformatics and Neuroscience. Final Grade: 1.2 (GPA 3.8)

Thesis: "Improving Physical Consistency in ML-driven Radiative Transfer Parameterization for Climate Model Emulation" Supervisors: Dr. Chirstoph Irrgang (RKI, Berlin), Prof. Dr. Gordan Pipa (IKW, Osnabrück), Grade 1.0 (GPA 4.0)

A Levels

Jacob-Grimm-Schule

📅 August 2015 – July 2018 📍 Kassel, DE

Focus on English and Mathematics. Final Grade: 1.1 (GPA 3.9)
Receiver of the school's graduate distinction in mathematics and physics. First place with the German team in an international physics competition (AYPT 2016).

TEACHING

Lecture: Implementing Artificial Neural Networks with TensorFlow

University of Osnabrück

📅 WS 20/21 & WS 21/22 📍 Osnabrück, DE

Co-lectured an 8 ECTS Bachelor/Master course, covering fundamentals of ANNs through state-of-the-art models (e.g., Transformers). Designed lectures and assignments; taught under Prof. Michael Franke (20/21) and Prof. Gordon Pipa (21/22).

Lecture: Deep Reinforcement Learning

University of Osnabrück

📅 WS 20/21 📍 Osnabrück, DE

Tutored an intensive 8 ECTS block course, developing a framework for efficient RL implementation, revising content, producing teaching graphics, and providing coding support. Covered basic RL through trust-region methods; taught by Leon Schmid under Prof. Elia Bruni.

PRACTICAL EXPERIENCE

• Research Assistant

📅 since October '25 📍 Geosense Lab, Freiburg

Initiated a collaboration between the GeoSense and the AutoML Labs, developing generalized AutoML workflows for environmental data within a project on global plant functional trait mapping.

• Independent Research Assistant

📅 May '23 - July '24 📍 MILA, remote

As part of an interdisciplinary, international project at the intersection of causal machine learning and climate physics, I handled coding, academic writing, method development, and management tasks.

• Honorary Tutor

📅 since April '23 📍 SFN, Kassel

At Schülerforschungszentrum Nordhessen (SFN) I mentor students in programming, AI, and climate science. My protégé **Lilly Schwarz** won the 2024 national *Jugend Forscht* interdisciplinary prize with her deep-learning climate-simulation project and is competing in 2026 with a deep-learning model for oceanic currents.

• Self-organized Research Internship

📅 Aug - Dec '22 📍 MILA, Montreal

During my stay at the Quebec Institute of AI, I worked on acquiring and preprocessing large-scale climate model data for machine-learning purposes in the context of a joint research project by MILA and Intel. Under Supervision of Prof. David Rolnick.

• Self-organized Research Internship

📅 June - July '22 📍 GFZ, Potsdam

During my stay at the Helmholtz Institute for Geophysics in the Geodesy Department (Section 1.3. Earth System Modelling) I explored their research projects and the prospects of applying machine learning. Supervised by Dr. Christoph Irrgang.

• DAAD funded Research Internship

📅 Aug - Nov '21 📍 UKZN, Pietermaritzburg

Developing an automated dataset collection system and prototyping a deep-learning based individual recognition software. Supervised by Dr. Mandlenkosi Gwetu.

PUBLICATIONS

Kaltenborn, J., Lange, C., Ramesh, V., Brouillar, P., Gurwicz, Y., Runge, J., Nowack, P., & Rolnick, D. (2023). ClimateSet: A Large-Scale Climate Model Dataset for Machine Learning. *Proceedings of the Thirty-Seventh Annual Conference on Neural Information Processing Systems, (NEURIPS 2023), New Orleans, USA, 10-16 December 2023.*

SKILLS

Python	●●●●●●●●
TensorFlow	●●●●●●●●
PyTorch	●●●●●●●●
end-to-end machine learning pipelines · distributed computing	
Linux (Ubuntu)	●●●●●●●●
setting up and configuring systems · virtual environments · problem-solving	
LaTeX	●●●●●●●●
R	●●●●●●●●
advanced statistics · time-series analysis	
Other	
basic SQL · basic GIS	

RESEARCH INTERESTS

theory-guided machine learning · automated machine learning (AutoML) · multimodal environmental data integration · forest & plant ecosystem modelling

LANGUAGES

German	●●●●●●●●
English	●●●●●●●●
French	●●●●●●●●

HOBBIES

exploring · poetry · creative writing · electronic music · nutrition-science · hiking

FURTHER EDUCATION

- International Summer School on Inventory and evaluation in nature conservation
📅 August 2025 📍 Romania
Grade: TBD
- International Summer School on Renewable Energies
📅 August 2019 📍 FH Aachen, Jülich
Grade: 2.0 (GPA 3.0)

COMMITMENT

University

- Student rep. for the appointment committee for the prof. in Neuromorphic Computing
📅 Jan - Aug 2022
- Member of the Institute of Cognitive Science20s Mentoring Team
📅 Aug 2020 - May 2022
- Student rep. for the appointment committee for the prof. in Machine Learning
📅 June 2020 - June 2021
- Elected vice- and president of the student parliament as a member of the Green Party
📅 May 2019 - October 2020